NARAYANA IIT ACADEMY INDIA

40+ YEARS OF EXCELLENCE

Sec: SR.IIT_*CO-SC(MODEL-A,B&C) | Date:21-01-24

Time: 3HRS | GTM-23(N)_FN | Max. Marks: 300

Name of the Student: ___________________ H.T. NO: [| | | | | | |]

21-01-24_SR.STAR CO-SUPER CHAINA(MODEL-A,B&C)_JEE-MAIN_GTM-23(N)-FN_SYLLABUS

PHYSICS: TOTAL SYLLABUS

CHEMISTRY: TOTAL SYLLABUS

MATHEMATICS: TOTAL SYLLABUS

SUBJECT	MISTAKES		
	JEE SYLLABUS Q'S	JEE EXTRA SYLLABUS Q'S	TOTAL Q'S
MATHS			
PHYISCS			
CHEMISTRY			

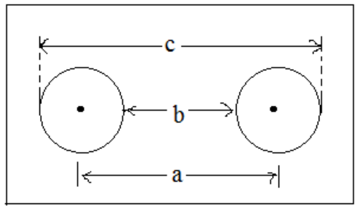
PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. A metal sheet (whose α is positive) has two circular holes as shown in figure. a is distance between centres of the holes and b is distance between nearest edges of the holes and c is distance between farthest edges of the holes as shown in figure. On heating



A) a remains constant b and c increase
B) a remains constant b decreases and c increase
C) All distances a,b,c increase D) a remains constant, b increases, c decreases

2. If the energy associated with electric field is (U_E) and with magnetic field is (U_B) for an electromagnetic wave in free space. Then:

A) $U_E = \frac{U_B}{2}$ B) $U_E > U_B$ C) $U_E < U_B$ D) $U_E = U_B$

3. A force acts on a 2 kg object so that its position is given as a function of time as $x = 3t^2 + 5$. What is the work done by this force in first 5 seconds?

A) 850 J B) 950 J C) 875 J D) 900 J

4. A particle having the same charge as of an electron moves in a circular path of radius 0.5 cm under the influence of a magnetic field of 0.5 T. If an electric field of 100 V/m makes it to move in a straight path, then the mass of the particle is (given charge of electron = 1.6×10^{-19} C)

A) 9.1×10^{-31} kg B) 1.6×10^{-27} kg C) 1.6×10^{-19} kg D) 2.0 × 10–24 kg

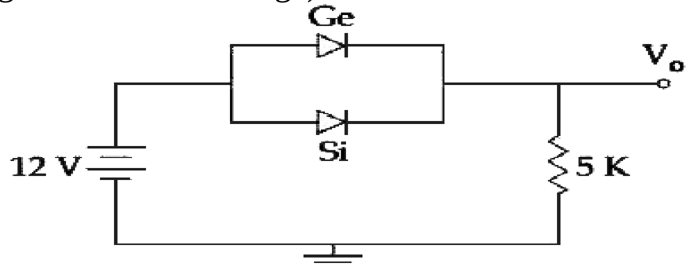
5. A particle executes simple harmonic motion with an amplitude of 10cm. If the angular frequency of oscillation of particle is 20 rad/s, the acceleration of the particle of its extreme position is

A) zero B) 100 m/s^2 C) 10 m/s^2 D) 40 m/s^2

6. Expression for time in terms of G (universal gravitational constant), h (Planck constant) and c (speed of light) is proportional to:

A) $\sqrt{\frac{hc^5}{G}}$ B) $\sqrt{\frac{c^3}{Gh}}$ C) $\sqrt{\frac{Gh}{c^5}}$ D) $\sqrt{\frac{Gh}{c^3}}$

7. Ge and Si diodes start conducting at 0.3 V and 0.7 V respectively. In the following figure if Ge diode connection are reversed, the value of V_0 changes by: (assume that the Ge diode has large breakdown voltage)



A) 0.8 V B) 0.6 V C) 0.2 V D) 0.4 V

8. The top of a water tank is open to air and its water level is maintained. It is giving out 0.74 m^3 water per minute through a circular opening of 2 cm radius in its wall. The depth of the centre of the opening from the level of water in the tank is close to:

A) 6.0 m B) 4.8 m C) 9.6 m D) 2.9 m

9. The energy required to take a satellite to a height 'h' above Earth surface (radius of Earth = 6.4×10^3 km) is E_1 and kinetic energy required for the satellite to be in a circular orbit at this height is E_2. The value of h for which E_1 and E_2 are equal is

A) 1.6×10^3 km B) 3.2×10^3 km C) 6.4×10^3 km D) 1.28×10^4 km

10. A series AC circuit containing an inductor (20 mH), a capacitor (120 μF) and a resistor (60 Ω) is driven by an AC source of 24 V/50 Hz. The energy dissipated in the circuit in 60 s is:

A) 5.65×10^2 J B) 2.26×10^3 J C) 5.17×10^2 J D) 3.39×10^3 J

11. A particle is executing simple harmonic motion (SHM) of amplitude A, along the x-axis, about x = 0. When its potential energy (PE) equals kinetic energy (KE), the position of the particle will be

A) $\frac{A}{2}$ B) $\frac{A}{2\sqrt{2}}$ C) $\frac{A}{\sqrt{2}}$ D) A

12. A mass of 10 kg is suspended vertically by a rope from the roof. When a horizontal force is applied on the rope at some point, the rope deviated at an angle of 45° at the roof point. If the suspended mass is at equilibrium, the magnitude of the force applied is ($g = 10\ ms^{-2}$)

A) 200 N B) 140 N C) 70 N D) 100 N

13. A 15 g mass of nitrogen gas is enclosed in a vessel at a temperature 27°C. Amount of heat transferred to the gas, so that rms velocity of molecules is doubled, is about: [Take R = 8.3 J/K mole]

A) 0.9 kJ B) 6 kJ C) 10 kJ D) 14 kJ

14. An object of mass m is projected with momentum P at such an angle that is its maximum height (H) is 1/4th of its horizontal range (R). Its minimum kinetic energy in its path will be

A) $\frac{p^2}{8m}$ B) $\frac{p^2}{4m}$ C) $\frac{3p^2}{4m}$ D) $\frac{p^2}{m}$

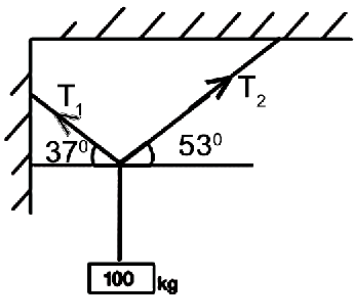
15. Two plane mirrors are inclined to each other such that a ray of light incident on the first mirror (M_1) and parallel to the second mirror (M_2) is finally reflected from the second mirror (M_2) parallel to the first mirror (M_1). The angle between the two mirrors will be:

A) 45° B) 60° C) 75° D) 90°

16. One of the two identical conducing wires of length L is bent in the form of a circular loop and the other one into a circular coil of N identical turns. If the same current is passed in both, the ratio of the magnetic field at the central of the loop (B_L) to that at the centre of the coil (B_C), i.e. $\frac{B_L}{B_C}$ will be

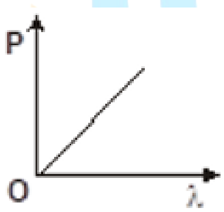
A) N B) $\frac{1}{N}$ C) N^2 D) $\frac{1}{N^2}$

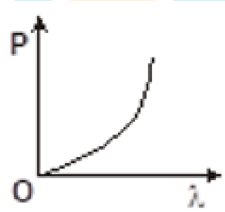
17. The object in figure weighs 100 kg and hangs at rest. Then T_1/T_2 is

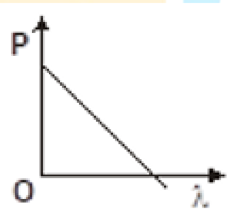


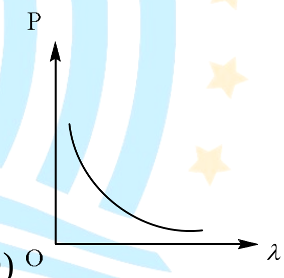
A) $\frac{4}{3}$ B) $\frac{3}{4}$ C) $\frac{9}{5}$ D) $\frac{5}{9}$

18. Which of the following figures best represents the variation of particle momentum and associated de Broglie wavelength?

A) 

B) 

C) 

D) 

19. Which of the following functions of time represents simple harmonic motion?

A) $x = \sin \omega t - \cos \omega t$ B) $x = \sin^3 \omega t$ C) $x = 1 + \omega t + \omega^2 t$ D) All of these

20. Radius of gyration of a body depends upon

A) Shape of the body B) Axis of rotation

C) Both (1) & (2) D) We can't say

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

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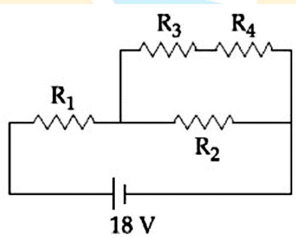
21. A rod of mass 'M' and length '2L' is suspended at its middle by a wire. It exhibits torsional oscillations; If two masses each of 'm' are attached at distance 'L/2' from its centre on both sides, it reduces the oscillation frequency by 20%. The value of ratio m/M is close to 0.ab then a+b is

22. The pitch and the number of divisions, on the circular scale, for a given screw gauge are 0.5 mm and 100 respectively. When the screw gauge is fully tightened without any object, the zero of its circular scale lies 3 divisions below the mean line. The readings of the main scale and the circular scale for a thin sheet, are 5.5 mm and 48 respectively, the thickness of this sheet is $5.abc$ mm then $\frac{a+c}{b}$ is

23. In a car race on straight road, car A takes a time 't=1.5 sec' less than car B at the finish and passes finishing point with a speed 'v' more than that of car B. Both the cars start from rest and travel with constant acceleration $a_1 = 9$ and $a_2 = 4$ respectively. Then 'v' is equal to

24. The magnetic field associated with a light wave is given, at the origin, by
$$B = B_0\left[\sin\left(3.14\times10^7\right)ct + \sin\left(6.28\times10^7\right)ct\right]$$
If this light falls on a silver plate having a work function of 4.7 eV, then maximum kinetic energy of the photo electrons is $a.bc\,eV$ then $b+c-a$ is

25. In the given circuit the internal resistance of the 18 V cell is negligible. If $R_1 = 400\,\Omega$, $R_3 = 100\,\Omega$ and $R_4 = 500\,\Omega$ and the reading of an ideal voltmeter across R_4 is 5 V, then the value of R_2 will be $50n$ then n is



26. The position co-ordinates of a particle moving in a 3-D coordinates system is given by
$x = a\cos\omega t$ $\quad y = a\sin\omega t$ $\quad$ and $\quad z = a\omega t$
The speed of the particle if $a = 2\sqrt{2}, \omega = 2$ is

27. A motor car is at a speed 60 m/s on a circular road of radius 1200m, which speeds up at a constant rate 4 m/s^2. At this instant the net acceleration of the car is (in m/s^2)

28. The radii of circular orbits of two satellites A and B of earth are 4R and R, respectively. If the speed of the satellite A is 3V, then the speed of satellite B will be _ __times that of V.

29. The power radiated by a black body is 'P', and it radiates maximum energy around the wavelength λ_0. If the temperature of the black body is now changed so that it radiates maximum energy around a wavelength $\lambda_0 / 2$, the power radiated by it will increase by a factor of

30. A capacitor stores 50μC charge when connected across a battery. When the gap between the plates is filled with a dielectric, a charge of 100 μC flows through the battery. The dielectric constant of the material is

CHEMISTRY MAX.MARKS: 100

CHEMISTRY
(SINGLE CORRECT ANSWER TYPE)

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31. Find the energy of the photons of radiation whose wavelength is $5000A^0$

A) $8.71.6\times10^{-16}J$ B) $4.35\times10^{-10}J$ C) $3.97\times10^{-10}J$ D) $4.97\times10^{-10}J$

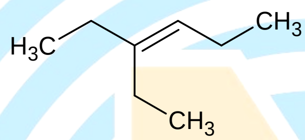
32. The total volume of mixture of 2 g of helium and 7 g of nitrogen under N.T.P conditions is

A) 22.4 lit B) 11.2 lit C) 16.8 lit D) 5.6 lit

33. Number of non-bonding electron pairs on Xe in XeF_6, XeF_4 and XeF_2 respectively will be

A) 6, 4, 2 B) 1, 2, 3 C) 3, 2, 1 D) 0, 3, 2

34. The correct IUPAC name of the following organic molecule is



A) 3-Ethylhex-4-ene B) 4-Ethylhex-4-ene

C) 3-Ethylhex-3-ene D) 4-Ethylhex-3-ene

35. Which one of the following is a monodentate ligand?

A) Chlorido B) Oxalato C) $EDTA^{-4}$ D) Ethylene diamine

36. Which one of the following does not undergo hydrolysis ?

A) NH_4Cl B) CH_3COONa C) $NaCl$ D) CH_3COONH_4

37. $SiCl_4$ is easily hydrolysed but CCl_4 is not. This is because

A) Bonding in $SiCl_4$ is ionic B) Siliconis non-metallic

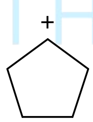
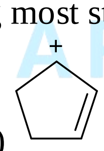
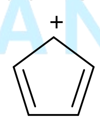
C) Silicon can extend its coordination number beyond four but carbon cannot

D) Silicon can form hydrogen bonds but carbon

38. $N_2(g)+3H_2(g)\rightleftharpoons 2NH_3(g)+energy$. The $\Delta H-\Delta E=$

A) $-2RT$ B) $+2RT$ C) $-RT$ D) $+RT$

39. Among the following most stable carbocation is

A)  B)  C)  D) 

40. The magnetic moment of the complex $\left[Ti(H_2O)_6\right]^{+3}$ is

A) 3.87 BM B) 1.73 BM C) 2.84 BM D) 5.87 BM

41. E^0 of $Mg^{2+}|Mg, Zn^{2+}|Zn$ and $Fe^{2+}|Fe$ are -2.37V, -0.76V and -0.44V respectively. Which of the following is correct?

A) Mg oxidizes Fe B) Zn reduces Fe^{2+} C) Zn reduces Mg^{2+} D) Zn oxidizes Fe

42. The major product(s) will be formed when but-1-yne upon reaction with dilute H_2SO_4 in the presence of Hg^{2+} ion as a catalyst

A) $CH_2-CH_2-CH_2-\overset{\overset{\displaystyle O}{||}}{C}-H$ B) $CH_3-CH_2-\overset{\overset{\displaystyle O}{||}}{C}-CH_3$

C) $CH_3-CH_2-\overset{\overset{\displaystyle O}{||}}{C}-OH+CO_2$ D) $CH_3-\overset{\overset{\displaystyle O}{||}}{C}-OH+H-\overset{\overset{\displaystyle O}{||}}{C}-H$

43. $R-C\equiv CH \xrightarrow[ii)CH_3CH_2Cl]{i)\,NaNH_2} product$. The product is

A) $R-C\equiv C-R$ B) $R-C\equiv R-CH_3$

C) $R-C\equiv C-CH_2CH_3$ D) $RCH=CH-CH_2CH_3$

44. Which among the following is very strong o-, p- directing group?

A) $-Cl$ B) $-OR$ C) $-NH_2$ D) $-O^-$

45. Which of the following will give yellow precipitate with $I_2/NaOH$

A) CH_3COCl B) CH_3COOCH_3 C) CH_3CONH_2 D) $CH_3CH(OH)CH_2CH_3$

46. The linkage between the two mono saccharides units in lactose is

A) C_1 of $\beta-D-Glucose$ & C_4 of $\beta-D-Fructose$

B) C_1 of $\beta-D-Galactose$ & C_4 of $\beta-D-Glucose$

C) C_1 of $\alpha-D-Fructose$ & C_4 of $\beta-D-Glucose$

D) C_1 of $\alpha-D-Galactose$ & C_4 of $\alpha-D-Fructose$

47. An unsaturated hydrocarbon on ozonolysis gives one mole each of methanal, ethanal and 2–keto propanal. The probable structure of the hydrocarbon is

A) $CH_2=CH-CH=CH-CH_3$ B) $CH_3-CH=CH-CH=CH-CH_3$

C) $CH_2=C(CH_3)-CH=CH-CH_3$ D) $CH_2=CH-CH(CH_3)-CH=CH_2$

48. Identify x and y in the following reaction

$$CH_3COOH+NH_3\rightarrow x\xrightarrow[-H_2O]{\Delta}y$$

A) CH_3CONH_2, CH_4 B) CH_3COONH_4, CH_3CONH_2

C) CH_3CONH_2, CH_3COOH D) CH_3-NH_2, CH_3CONH_2

49. Which of the following is a diamagnetic complex

A) $K_4[Fe(CN)_6]$ B) $[NiCl_4]^{-2}$ C) $K_3[Fe(CN)_6]$ D) $[Fe(H_2O)_6]Cl_3$

50. The number of electrons required to deposit one mole of Mg^{+2} ions

A) 6.023×10^{23} B) 12.046×10^{23} C) 18.069×10^{23} D) 3.012×10^{23}

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

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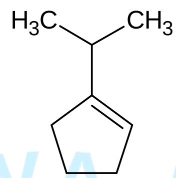
51. The rate of gaseous reaction is given by $k[A][B]$. If the volume of reaction vessels is reduced

to $\frac{1}{4}$th of initial volume. The reaction rate relative to the original rate is

52. The maximum number of carbon atoms in the major product of the following reaction is

$$CH_3CH_2CH_2Cl \xrightarrow[\text{Dry Ether}]{Na} \text{Major product}$$

53. The maximum number of sigma C-H bonds involved in hyperconjugation in the given compound is



54. The vapour pressure of pure solvent is 100 mm of Hg. When a non volatile solute is added to it, the vapour pressure becomes 80 mm of Hg. If the RLVP of solution is $y \times 10^{-1}$ then y is

55. The p^H of solution is 4, if it is diluted 100 times. Then the p^H of the solution is

56. The number of moles of MnO_4^- (In acidic medium) required to oxidize 50 ml 0.1 M aq. Fe^{+2} solution is ____$\times 10^{-3}$

57. How many of the following compounds violet octet rule ?

a) BrF_5 b) SF_6 c) IF_7 d) $XeOF_4$ e) NH_3 f) PCl_5

58. The total number of nodes in $4P_x$ orbital are

59. Some of the group number and period number of an element of atomic number 50 is (According to IUPAC)

60. The molarity of 11.2 V H_2O_2 is

MATHEMATICS MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

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61. A fair coin is tossed 5 times. Then the probability that no two heads occur consecutively is

A) $\frac{1}{16}$ B) $\frac{15}{32}$ C) $\frac{13}{32}$ D) $\frac{7}{16}$

62. If the roots of the equation $x^2-(k+1)x+(k^2+k-8)=0$ are on either side of 2 then the value of k can be

A) 4 B) -4 C) 2 D) -2

63. The equation of the line through the point (1, 2, 3) and parallel to the line $\frac{x-4}{2}=\frac{y-1}{-3}=\frac{z+10}{8}$ is

A) $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{8}$ B) $\frac{x-1}{2}=\frac{y-2}{-3}=\frac{z-3}{8}$

C) $\frac{x-4}{1}=\frac{y-1}{2}=\frac{z+10}{3}$ D) $\frac{x-4}{2}=\frac{y-2}{-3}=\frac{z+10}{3}$

64. The angle between the lines whose direction cosines satisfy the equation $l+m+n=0, l^2+m^2-n^2=0$ is given by

A) $2\pi/3$ B) $\pi/6$ C) $5\pi/6$ D) $\pi/3$

65. The area of the region bounded by the parabola $y^2=4x$ and its latusrectum is

A) 1 B) ¾ C) 8/3 D) 4/3

66. $\int \frac{e^x(1+\sin x)}{1+\cos x}dx =$

A) $\log|\tan x|+c$ B) $e^x \tan x/2+c$ C) $\sin(e^x).\cot x+c$ D) $e^x \cot x+c$

67. If $x=3\tan t$ and $y=3\sec t$, then the value of $\frac{d^2y}{dx^2}$ at $t=\frac{\pi}{4}$, is

A) $\frac{3}{2\sqrt{2}}$ B) $\frac{1}{3\sqrt{2}}$ C) $\frac{1}{6}$ D) $\frac{1}{6\sqrt{2}}$

68. The number of natural numbers less than 7,000 which can be formed by using the digits 0, 1, 3, 7, 9 (repetition of digits allowed) is equal to

A) 250 B) 374 C) 372 D) 375

69. If $\mathbf{a}=2\mathbf{i}+\mathbf{k}$, $\mathbf{b}=\mathbf{i}+\mathbf{j}+\mathbf{k}$ and $\mathbf{c}=4\mathbf{i}-3\mathbf{j}+7\mathbf{k}$. If $\mathbf{d}\times\mathbf{b}=\mathbf{c}\times\mathbf{b}$ and $\mathbf{d}.\mathbf{a}=0$, then d will be

A) $\mathbf{i}+8\mathbf{j}+2\mathbf{k}$ B) $\mathbf{i}-8\mathbf{j}+2\mathbf{k}$ C) $-\mathbf{i}+8\mathbf{j}-\mathbf{k}$ D) $-\mathbf{i}-8\mathbf{j}+2\mathbf{k}$

70. If the real valued function $f(x)=\frac{a^x-1}{x^n(a^x+1)}$ is even, than n=

A) 6 B) 4 C) 3 D) 12

71. If the sum of an infinite GP is $\frac{7}{2}$ and sum of the squares of its terms is $\frac{147}{16}$, then the sum of the cubes of its terms is

A) $\frac{315}{19}$ B) $\frac{700}{39}$ C) $\frac{985}{13}$ D) $\frac{1029}{38}$

72. The eccentricity of a hyperbola, whose conjugate hyperbola is $2x^2-y^2=6$ is

A) $\frac{1}{\sqrt{2}}$ B) $\sqrt{2}$ C) $\sqrt{3}$ D) $\sqrt{\frac{3}{2}}$

73. If $3A+4B^T=\begin{pmatrix}7 & -10 & 17\\ 0 & 6 & 31\end{pmatrix}$ and $2B-3A^T=\begin{pmatrix}-1 & 18\\ 4 & -6\\ -5 & -7\end{pmatrix}$ then B=

A) $\begin{pmatrix}1 & 3\\ -1 & 0\\ -2 & -4\end{pmatrix}$ B) $\begin{pmatrix}1 & 3\\ 1 & 0\\ 2 & 4\end{pmatrix}$ C) $\begin{pmatrix}1 & 3\\ -1 & 0\\ 2 & 4\end{pmatrix}$ D) $\begin{pmatrix}1 & -3\\ 1 & 0\\ 2 & 4\end{pmatrix}$

74. Let $A=\begin{pmatrix}5 & 5\alpha & \alpha\\ 0 & \alpha & 5\alpha\\ 0 & 0 & 5\end{pmatrix}$. If $|A^2|=25$, then $|\alpha|=$

A) $\frac{1}{5}$ B) 5 C) 5^2 D) 1

75. A random variable X has the following distribution

$X = x_i$	1	2	3	4
$P(X = x_i)$	K	2K	3K	4K

The value of 'K' and P(X<3) are respectively equal to

A) $\frac{1}{10}, \frac{3}{5}$ B) $\frac{1}{10}, \frac{3}{10}$ C) $\frac{3}{10}, \frac{1}{10}$ D) $\frac{1}{24}, \frac{5}{12}$

76. Let $A = \{x \in R : -1 \le x \le 1\} = B$. Then the mapping $f : A \to B$ given by $f(x) = x|x|$ is

A) Injective but not surjective B) Surjective but not injective

C) Bijective D) None of these

77. The system $2x + 3y + z = 5,\ 3x + y + 5z = 7,\ x + 4y - 2z = 3$ has

A) Unique solution B) Finite number of solutions

C) Infinite solutions D) No solution

78. The value of 'k' so that the lines $\frac{x+3}{-3} = \frac{y-1}{1} = \frac{z-5}{5}$ and $\frac{x+1}{-1} = \frac{y-2}{k} = \frac{z-5}{5}$ are coplanar, is

A) 1 B) 0 C) 2 D) -2

79. For two data sets, each of size 5, the variances are given to be 4 and 5 and the corresponding means are given to be 2 and 4 respectively. The variance of the combined data set is

A) $\frac{5}{2}$ B) $\frac{11}{2}$ C) 6 D) $\frac{13}{2}$

80. $\tan^{-1} 1 + \tan^{-1} 2 + \tan^{-1} 3 =$ ______

A) $\frac{\pi}{2}$ B) π C) $3\frac{\pi}{2}$ D) 2π

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, -1 in all other cases.

81. A hyperbola has its centre at the origin, passes through the point (4, 2) and has transverse axis of length 4 along the x – axis. Then the eccentricity of the hyperbola is

82. If $x = \sin^{-1}(\sin 10)$ and $y = \cos^{-1}(\cos 10)$, then $y - x$ is equal to

83. If $z_1 = 10 + 6i, z_2 = 4 + 6i$ and z is a complex number such that $amp\left(\frac{z - z_1}{z - z_2}\right) = \frac{\pi}{4}$, then the value of $|z - 7 - 9i|$ is equal to

84. If a straight line through $C(-\sqrt{8}, \sqrt{8})$ making an angle of $135°$ with the *x*-axis cuts the circle $x = 5\cos\theta, y = 5\sin\theta$ at points *A* and *B*, then the length of *AB* is

85. $\lim_{x \to 0} \frac{1 - \cos 2x}{x^2}$ is

86. If $\left|z - \frac{3}{z}\right| = 2$, then the greatest value of $|z|$ is

87. If $A = \{1, 2, 3, 4, 5\}$, then the number of subsets of A is_____

88. Sum of the coefficients in the expansion of $(2x+3y+2z-5t)^{10}$ is ______

89. A man has 7 relatives of which 4 are women and 3 are men. His wife also has 7 relatives of which 3 are women and 4 are men. The number of ways in which they can invite 3 men and 3 women so that they both invite three each is…..

90. The degree of the differential equation of $y=(c_1+c_2)\cos(x+c_3)-c_4e^{x+c_5}$ where $c_1, c_2 ....c_5$ are arbitrary constants is